

Scattering by a Central Potential; The Hamiltonian and Hamilton's Equations of Motion

HC 6001 — Dr. Castillo

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This chapter typesets the instructor's handwritten lecture notes for HC 6001 (October 22, 2024) in the notation and equation conventions of Goldstein, Poole & Safko, *Classical Mechanics*, 3rd ed. Two topics are covered: the kinematics and dynamics of scattering by a central potential, culminating in the Rutherford cross section (Sections 1–3), and the passage from the Lagrangian to the Hamiltonian formalism via the Legendre transform, the opening of the course's Chapter 8 (Section 4). Occasional remarks below that refer to “the original notes” point to the instructor's handwritten source pages, reproduced in part as figures throughout.

1 Kinematics of Scattering by a Central Potential

Consider a particle of mass m incident on a fixed central-potential scattering center with energy E , velocity v_0 , and angular momentum ℓ :

$$E = \frac{m}{2}v_0^2, \quad v_0 = \sqrt{2E/m}, \quad \ell = smv_0 = s\sqrt{2mE}, \quad (1)$$

where s is the *impact parameter*. Equivalently, $s = \ell/\sqrt{2mE}$.

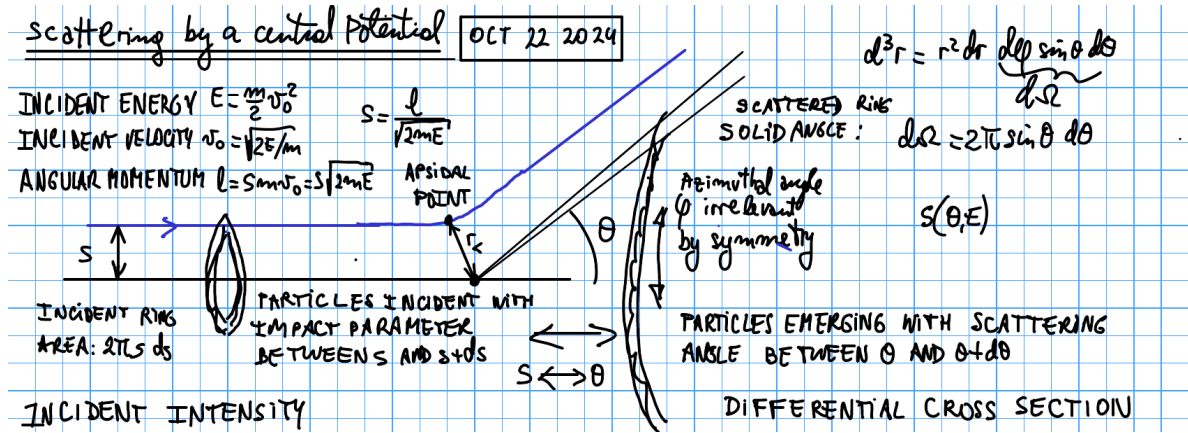


Figure 1: Impact-parameter and scattered-ring geometry: particles incident with impact parameter between s and $s + ds$ (left) emerge with scattering angle between θ and $\theta + d\theta$ into the ring of solid angle $d\Omega$ (right). The azimuthal angle φ is irrelevant by the symmetry of a central potential, so the scattering problem reduces to a function $s = S(\theta, E)$.

As Fig. 1 also records, the solid-angle element used throughout is $d^3r = r^2 dr d\varphi \sin \theta d\theta$, with $d\Omega \equiv d\varphi \sin \theta d\theta$ integrated over φ to give $d\Omega = 2\pi \sin \theta d\theta$ below.

$$I = \frac{\text{\# particles incident per unit time}}{\text{incident beam cross-section area}}.$$
$$I(2\pi s) |ds| = I \sigma(\theta) 2\pi \sin \theta |d\theta|,$$
$$\left| \sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| \right|, \quad \boxed{\sigma_T = \int d\Omega \, \sigma(\Omega)}, \quad (2)$$

REPUSSIVE CASE:

$$\pi = \theta + 2\psi$$

$$\theta = \pi - 2\psi$$

$$\psi = \frac{\pi - \theta}{2}$$

IN ALL CASES:

$$\theta = |\pi - 2\psi|$$

ATTRACTIVE CASE:

$$\pi = 2\psi - \theta$$

$$\theta = 2\psi - \pi$$

$$\psi = \frac{\pi + \theta}{2}$$

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$\text{Repulsive: } \pi = \theta + 2\psi, \quad \theta = \pi - 2\psi, \quad \psi = \frac{\pi - \theta}{2}; \quad (3)$$

$$\theta = |\pi - 2\psi|. \quad (5)$$

$$d\theta = \dot{\theta} dt = \frac{\ell}{mr^2} \frac{dr}{\dot{r}} = \frac{s r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \frac{s^2}{r^2}\right]^{1/2}},$$

using $\ell/\sqrt{2mE} = s$. Integrating from the apsidal distance r_m (where $u_m \equiv 1/r_m$) out to $r \rightarrow \infty$, and substituting $u = 1/r$,

$$\psi = \int_0^{u_m} \frac{s du}{\left[1 - \frac{V(1/u)}{E} - s^2 u^2\right]^{1/2}}, \quad (6)$$

so that, combined with Eq. (5),

$$\theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s du}{\left[1 - \frac{V(1/u)}{E} - s^2 u^2\right]^{1/2}} \right|. \quad (7)$$

From $\theta(s)$ one computes $|ds/d\theta| = 1/|d\theta/ds|$ and hence $\sigma(\theta)$ via Eq. (2).

Remark 1. Equations (5)–(7) are valid for *any* central potential $V(r)$; nothing in this section has assumed a specific force law.

3 Coulomb (Rutherford) Scattering

We now specialize the universal result Eq. (7) to the Coulomb interaction between two charges Ze and $Z'e$ (Gaussian units, following Goldstein's convention), with force $f = ZZ'e^2/r^2 = -k/r^2$ and potential $V(r) = -k/r$, $k = -ZZ'e^2$.

For $V(r) = -k/r$ the orbit is a conic section,

$$\frac{1}{r} = C_1 [1 + \varepsilon \cos(\tilde{\theta} - \tilde{\theta}')], \quad C_1 = \frac{mk}{\ell^2}, \quad \varepsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1 \quad (8)$$

for $E > 0$, using $\ell = s\sqrt{2mE}$.

Remark 2 (Orbit classification). For this potential the sign of E fixes the orbit type: $E > 0$ gives a hyperbolic orbit ($\varepsilon > 1$), $E = 0$ a parabolic orbit ($\varepsilon = 1$), and $E < 0$ an elliptic orbit ($\varepsilon < 1$). This three-way classification by the sign of E is specific to $V(r) = -k/r$ and does not hold for a general central potential. Scattering states have $E > 0$, so only the hyperbolic case is used below.

Two situations occur:

$$\text{repulsive: } ZZ' > 0 \Rightarrow k = -ZZ'e^2 < 0, \quad C_1 < 0; \quad (9)$$

$$\text{attractive: } ZZ' < 0 \Rightarrow k = -ZZ'e^2 > 0, \quad C_1 > 0 \quad (\text{also gravity, } k = Gm_1m_2). \quad (10)$$

Repulsive case ($k < 0$). Physical requires $1/r > 0$, i.e. $\varepsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1$; since $\tilde{\theta} = \tilde{\theta}'$ is unphysical (it would give $1/r = C_1(1 + \varepsilon) < 0$), the periapsis is at $\tilde{\theta}' = \pi$, so $\cos(\tilde{\theta} - \tilde{\theta}') = -\cos \tilde{\theta}$ and

$$\frac{1}{r} = \frac{mZZ'e^2}{\ell^2} [\varepsilon \cos \tilde{\theta} - 1],$$

with apsidal angle $\tilde{\theta} = 0$. The distance of closest approach (minimum r , maximum $1/r$) therefore occurs at $\tilde{\theta} = 0$, giving $1/r_m = (mZZ'e^2/\ell^2)(\varepsilon - 1)$. Requiring $\psi = \tilde{\theta}(r \rightarrow \infty)$, i.e. $1/r \rightarrow 0$, gives $\varepsilon \cos \psi = 1$, and using $\theta = \pi - 2\psi$ (repulsive case) one finds $\frac{1}{\varepsilon} = \sin \frac{\theta}{2}$.

Attractive case ($k > 0$). By symmetry ($\tilde{\theta}' = 0$ is now the periapsis), the minimum distance is $1/r_m = C_1(1 + \varepsilon)$, again at $\tilde{\theta} = 0$. The same asymptotic analysis with $\theta = 2\psi - \pi$ gives $-\frac{1}{\varepsilon} = -\sin \frac{\theta}{2}$, i.e. the same relation $\frac{1}{\varepsilon} = \sin \frac{\theta}{2}$.

Combining both cases. Since $\cot^2(\theta/2) = 1/\sin^2(\theta/2) - 1 = \varepsilon^2 - 1$, and from Eq. (8), $\varepsilon^2 - 1 = (2sE/ZZ'e^2)^2$, one obtains the impact parameter as a function of scattering angle,

$$s = S(\theta, E) = \left| \frac{ZZ'e^2}{2E} \right| \cot \frac{\theta}{2}. \quad (11)$$

Differentiating,

$$\frac{ds}{d\theta} = -\frac{1}{2} \left| \frac{ZZ'e^2}{2E} \right| \csc^2 \frac{\theta}{2},$$

and substituting s and $|ds/d\theta|$ into Eq. (2), using $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$,

$$\sigma(\theta) = \frac{1}{\sin \theta} \left| \frac{ZZ'e^2}{2E} \right|^2 \cot \frac{\theta}{2} \cdot \frac{1}{2} \csc^2 \frac{\theta}{2} = \left(\frac{ZZ'e^2}{2E} \right)^2 \frac{1}{4} \csc^4 \frac{\theta}{2},$$

i.e. the **Rutherford scattering cross section** for Coulomb forces:

$$\sigma(\theta) = \frac{1}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \csc^4 \frac{\theta}{2}. \quad (12)$$

Remark 3. The cross section (12) is the same for the repulsive and attractive cases, since it depends on ZZ' only through $(ZZ')^2$.

Remark 4. It diverges like E^{-2} at low energies.

Remark 5. It diverges like θ^{-4} (*very strong*) at small scattering angles.

Remark 6. This classical result is exactly the same as the result obtained in quantum mechanics.¹

Remark 7 (Notation). Following Goldstein's convention, the differential and total cross sections are written $\sigma(\Omega)$ and σ_T ; other references commonly write $d\sigma/d\Omega$ and σ for the same two quantities, respectively.

3.1 Total Scattering Cross Section

With $\sigma(\theta) = \frac{1}{4}(ZZ'e^2/2E)^2 \csc^4(\theta/2)$ and $d\Omega = 2\pi \sin \theta d\theta = 4\pi \sin(\theta/2) \cos(\theta/2) d\theta$,

$$\sigma_T = \frac{1}{4} \left(\frac{ZZ'e^2}{2E} \right)^2 \cdot 4\pi \int_0^\pi \frac{\sin(\theta/2) \cos(\theta/2)}{\sin^4(\theta/2)} d\theta = \pi \left(\frac{ZZ'e^2}{2E} \right)^2 \int_0^\pi \frac{\cos(\theta/2)}{\sin^3(\theta/2)} d\theta \sim \int_0 \frac{d\theta}{(\theta/2)^3},$$

which is **strongly divergent** as $\theta \rightarrow 0$: classically, $\sigma_T = +\infty$ for any long-range potential; quantum mechanically, $\sigma_T = +\infty$ as well for potentials $V(r)$ that do not decay fast enough as $r \rightarrow \infty$.

Caveat 1. In a dense, conducting material the bare Coulomb potential $V = -k/r$ is replaced by the screened ("Yukawa") potential

$$V^{\text{screen}}(r) = -\frac{k}{r} e^{-r/\lambda},$$

which regularizes the divergence. In non-conducting materials $V^{\text{screen}}(r)$ may take a different functional form, but will still vanish much faster than $1/r$ as $r \rightarrow \infty$.

¹Editorial note, not in the original notes: the first Born approximation for the Coulomb potential reproduces Eq. (12) exactly, which is the standard explanation for why the classical and quantum cross sections coincide here.

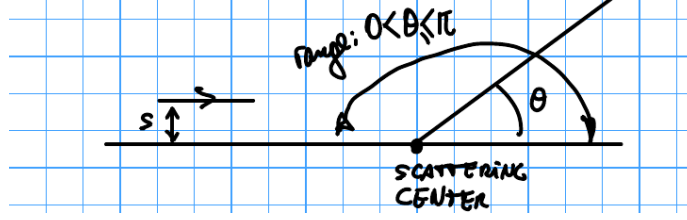


Figure 3: Scattering-center geometry: a particle with impact parameter between s and $s + ds$, incident along the horizontal line, is deflected by angle θ at the scattering center.

4 The Hamiltonian and Hamilton's Equations of Motion

We now turn to a different topic, which opens the course's Chapter 8 immediately following the scattering material of Sections 1–3 in the same lecture: rather than continuing with the Lagrangian description $L(q, \dot{q}, t)$ used implicitly above, we develop an equivalent formulation in terms of the generalized coordinates and their conjugate momenta, (q, p, t) . Starting from the Lagrangian $L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$ and the Euler–Lagrange equations

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i}, \quad p_i \equiv \left. \frac{\partial L}{\partial \dot{q}_i} \right|_{q_j, \dot{q}_j},$$

Remark 8 (Midterm remark — an invalid claim). In plane polar coordinates it is tempting to write $\partial L / \partial \dot{\theta} = p_\theta = \ell_z$ and conclude $L = L(r, \dot{r}, p_\theta)$, i.e. to substitute the canonical momentum p_θ directly back into the Lagrangian in place of $\dot{\theta}$. This step is **invalid**: the Lagrangian is a function of the generalized coordinates and *velocities*, $L = L(q, \dot{q}, t)$; replacing $\dot{q}$ by its conjugate momentum p changes the independent variables and is precisely the Legendre transformation carried out below to pass from L to H — it cannot be done term-by-term inside L itself. Consequently the crossed-out “ $\Rightarrow$ Lagrange Eqn.” that follows in the original notes is equally invalid: substituting the ill-defined $L(r, \dot{r}, p_\theta)$ into the Euler–Lagrange equation would silently discard the r -dependence hidden inside $p_\theta = mr^2 \dot{\theta}$. (Marked as crossed out in the original notes as a caution against this common exam error.)

4.1 The Legendre Transformation

The passage from $L(q, \dot{q}, t)$ to a new function of (q, p, t) is an instance of the general Legendre transformation. For $f(x, y)$ with $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \equiv u dx + v dy$, define $g = f - ux$; then

$$dg = df - d(ux) = u dx + v dy - u dx - x du = -x du + v dy,$$

so $g = g(u, y)$: the transformation trades the variable x for the conjugate variable $u = \partial f / \partial x$.

Remark 9 (Thermodynamic analogy). The same transformation appears in thermodynamics. With the first law $dU = \delta Q + \delta W = T dS - p dV$ (fluid at fixed particle number N), $U = U(S, V)$ is the internal energy. To use (T, V) as independent variables instead, define the Helmholtz free energy $A = U - TS$:

$$dA = dU - d(TS) = (T dS - p dV) - T dS - S dT = -S dT - p dV \Rightarrow A = A(T, V),$$

exactly the Legendre transform $g = f - ux$ with $(x, u) \rightarrow (S, T)$.

4.2 From the Lagrangian to the Hamiltonian

Apply the Legendre transform to $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$, trading each $\dot{q}_i$ for its conjugate momentum $p_i = \partial L / \partial \dot{q}_i$. Define

$$H \equiv -\left(L - \sum_i p_i \dot{q}_i\right) = \sum_i p_i \dot{q}_i - L.$$

Then

$$dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) - \frac{\partial L}{\partial t} dt + \sum_i (p_i d\dot{q}_i + \dot{q}_i dp_i),$$

and since $\partial L / \partial \dot{q}_i = p_i$ the $d\dot{q}_i$ terms cancel:

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \implies H = H(p_1, \dots, p_n, q_1, \dots, q_n, t).$$

Using the Euler–Lagrange equation, $\partial L / \partial q_i = \dot{p}_i$, and comparing the two expressions for dH term by term gives **Hamilton’s canonical equations of motion**:

$$\boxed{\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.} \quad (13)$$